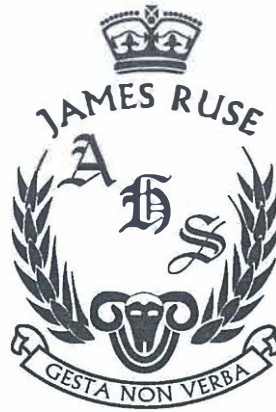


|        |  |
|--------|--|
| Name:  |  |
| Class: |  |



**YEAR 12**

**ASSESSMENT TEST 2**

**TERM 1, 2017**

**MATHEMATICS**

*Time Allowed – 120 Minutes  
(Plus 5 minutes Reading Time)*

*General Instructions:*

- All questions may be attempted
- All multiple choice questions are of equal value
- Reference Sheet will be supplied
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper

**The answers to all questions are to be returned in separate bundles clearly labelled Question 6, Question 7, etc. Each question must show your Candidate Number.**

ANSWER THE FOLLOWING QUESTIONS, ON YOUR MULTIPLE CHOICE ANSWER SHEET.

1. What is  $0.\dot{3}\dot{5}$  as a fraction in its simplest form?

A]  $\frac{7}{20}$

B]  $\frac{7}{18}$

C]  $\frac{35}{99}$

D]  $\frac{35}{100}$

2. Differentiate  $y = \sqrt[3]{\sin^2 6x}$

A]  $\frac{2\cos 6x}{3\sqrt[3]{\sin 6x}}$

B]  $\frac{4\cos^2 6x}{3\sqrt[3]{\sin^2 6x}}$

C]  $\frac{4\cos 6x}{3\sqrt[3]{\sin^2 6x}}$

D]  $\frac{4\cos 6x}{3\sqrt[3]{\sin 6x}}$

3. Evaluate exactly  $\int_1^2 \frac{2x+1}{x+x^2} dx$

A]  $\ln 3$

B]  $\ln 12$

C]  $\ln 6$

D]  $\ln 4$

4. Which line is perpendicular to  $3x + 4y + 7 = 0$ ?

A]  $4x + 3y - 7 = 0$

B]  $3x - 4y + 7 = 0$

C]  $8x - 6y - 7 = 0$

D]  $4x - 7y + 7 = 0$

5. For what values of  $m$  will the geometric series

$1 + 2m + 4m^2 + 8m^3 + \dots$  have a limiting sum?

A]  $-1 \leq m \leq 1$

B]  $\frac{-1}{2} \leq m \leq \frac{1}{2}$

C]  $\frac{-1}{2} < m < \frac{1}{2}$

D]  $m < \frac{1}{2}$

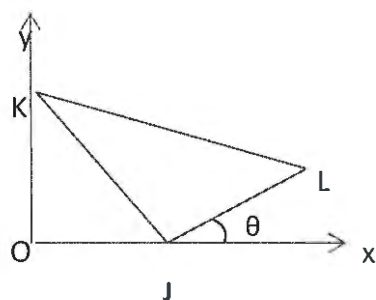
## Question Six (27 marks)

(a) Convert  $48^\circ$  to radians; give your answer as an exact value. 1

(b) The points J, K and L have coordinates (1, 0), (0,8) and (7,4).

The angle between the line JL and the x-axis is  $\theta$ .

Copy the diagram below neatly onto your answer sheet.



(i) Find the gradient of JL 1

(ii) Find the size of angle  $\theta$  to the nearest degree. 1

(iii) Find the coordinates of M, the midpoint of JL. 1

(iv) Show that JL is perpendicular to KM. 2

(v) Find the area of  $\triangle JKL$  2

(vi) Write down the coordinates of the point N which makes JKLN a rhombus. 2

(c) Is  $\frac{9}{2}$  a term of the sequence  $T_n = \frac{3}{32}(2^{n-2})$ ? Give reasons. 2

(d) (i) Find  $T_n$  for  $14 + 11 + 8 + 5 + \dots$  2

(ii) Express  $14 + 11 + 8 + 5 + \dots + -34$  using Sigma notation. 2

(e) For the following sequence find the general term in simplest form.

$\log_b 5x^2, \log_b 5x, \log_b 5, \dots$  3

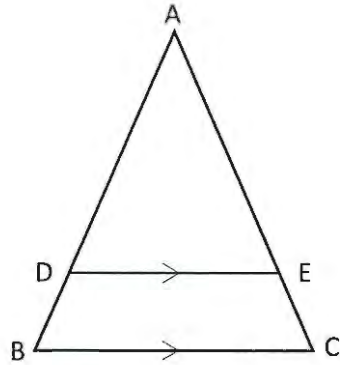
(f) If  $S_n = n^2 - 4n$  find an expression for  $T_n$ . 2

(g) In the diagram below;  $AE = 18\text{cm}$ ,  $EC = 6\text{cm}$ ,  $AB = 16\text{cm}$ .

**Copy** the diagram onto your answer page.

(i) Prove that  $\triangle ADE$  is similar to  $\triangle ABC$ . 2

(ii) Find the length of  $DB$  giving reasons. 2



(h) Find the size of the angles of the isosceles triangle in which the base angles are double the vertical angle. 2

### Question Seven (27 marks)

**START A NEW PAGE**

(a) Draw a possible graph of the curve that satisfies all of the following conditions  $0 \leq x \leq a$ ,  $y'' < 0$ ,  $y' > 0$  and  $y > 0$ . 1

(b) On the separate question paper provided, neatly graph  $y = f(x)$  and  $y = f''(x)$ .  
Clearly label **each** graph. 2

(c) (i) Find the coordinates of the stationary points and determine their nature for the function  $y = x^3 - 12x + 5$ . 4

(ii) Find the coordinates of any points of inflexion. 2

(iii) Neatly sketch  $y = x^3 - 12x + 5$  (no need to find x-intercepts). 1

- (d) The area of a sector AOB is  $96\text{cm}^2$  and its perimeter is 56cm. Find the length of the radius and the size of the angle of the sector to the nearest degree. (O is the centre of the circle). 5
- (e) The number 36 is divided into two parts. The smaller part is multiplied by the square of the larger part.
- (i) Show the product is  $P = 1296x - 72x^2 + x^3$  1
- (ii) Hence find the maximum product possible. 4
- (f) If  $\frac{dy}{dx} = \sec^2 2x$ , and  $y = 0$  when  $x = \frac{\pi}{6}$  find an expression for  $y$ . 2
- (g) Find the exact area enclosed by the curve  $y = 4\cos 2x$  from  $x = -\frac{\pi}{6}$  to  $x = \frac{\pi}{6}$ . 2
- (h) Find the volume of the solid of revolution when  $y = e^{x+1}$  is rotated about the  $x$ -axis from  $x = 0$  to  $x = \ln 3$  correct to two decimal places. 3

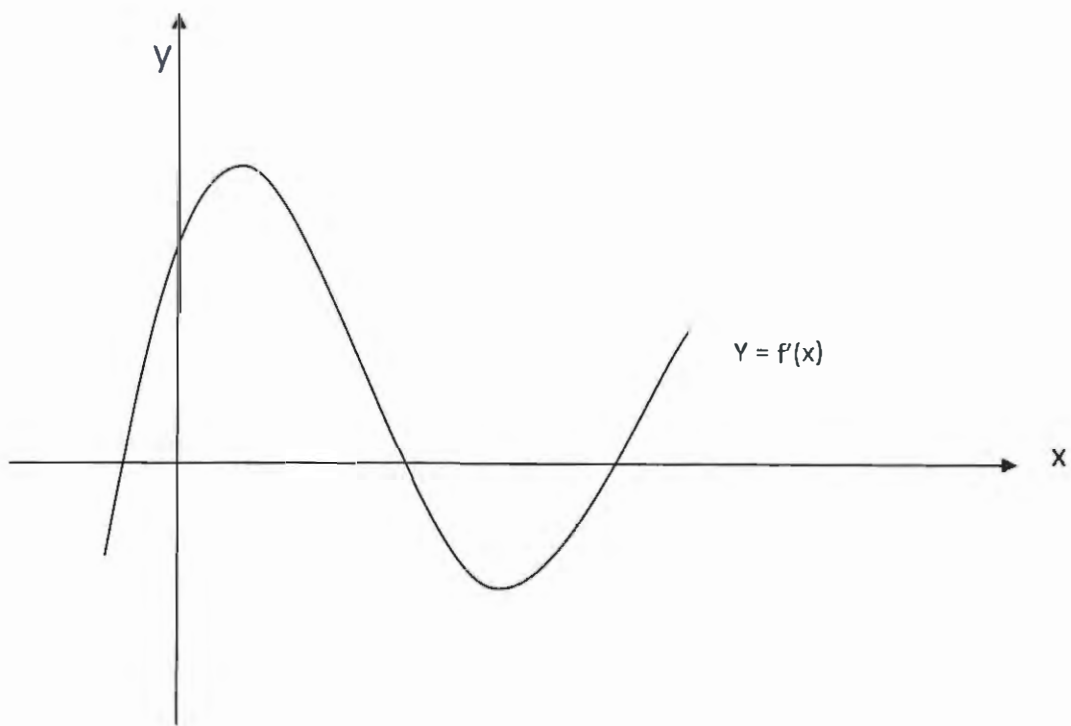
### Question Eight (27 marks)

**START A NEW PAGE**

- (a) (i) Show that the first derivative of  $y = \ln \sqrt{\frac{1+x}{1-x}}$  is  $\frac{dy}{dx} = \frac{1}{1-x^2}$ . 2
- (ii) Hence or otherwise evaluate exactly  $\int_0^{\frac{1}{3}} \frac{4dx}{1-x^2}$  2
- (b) Evaluate exactly  $\int_0^{\frac{\pi}{8}} (1 - \cot 4x) dx$  3
- (c) Show that the exact area bounded by the curve  $y = \ln 3x$ , the  $x$ -axis and the lines  $x = \frac{2}{3}$  and  $x = 1$  is given by  $A = \frac{1}{3} (3\ln 3 - 2\ln 2 - 1)$ . 3

- (d) Using Simpson's rule with five functional values approximate the volume of revolution when  $y = x \sin x$  is rotated about the  $x$ -axis from  $x = 0$  to  $x = \frac{\pi}{2}$ . (Express your answer correct to two decimal places). 4
- (e) A cylinder is inscribed in a sphere of radius 12cm.
- (i) If the height of the cylinder is  $h$ , show that the volume of the cylinder is given by  $V = \frac{\pi h}{4}(576 - h^2)$ . 2
- (ii) Find the maximum volume of the cylinder possible. 3
- (iii) Find the ratio of this greatest volume of the cylinder to the volume of the sphere. 1
- (f) (i) Graph neatly  $y = 3 - 2 \sin x$  for  $0 \leq x \leq 2\pi$ . 2
- (ii) How many solutions does the equation  $1 - \sin x - \frac{x}{3} = 0$  have for  $0 \leq x \leq 2\pi$ ? 2
- (g) Leonard invests \$10 000 into an interest bearing account. At the beginning of every month, starting one month after opening the account, he deposits \$ $m$ . Interest of 5% p.a. is compounded monthly and is paid at the end of the month. If after 5 years immediately after the interest has been paid his account balance is \$160 000, find the size of the monthly deposit. 3

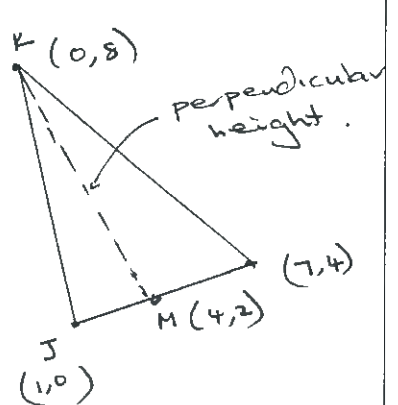
*End of Paper*



| MATHEMATICS: Question <i>m</i> choice                                                                                                                              |       |                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------|
| Suggested Solutions                                                                                                                                                | Marks | Marker's Comments |
| 1. $0.\dot{3}\dot{5} = \frac{35}{99}$                                                                                                                              | C     |                   |
| 2. $y = \sqrt[3]{\sin^2 6x}$<br>$y = (\sin 6x)^{2/3}$<br>$\frac{dy}{dx} = \frac{2}{3} (\sin 6x)^{-1/3} \cdot 6 \cos 6x$<br>$= \frac{4 \cos 6x}{\sqrt[3]{\sin 6x}}$ | D     |                   |
| 3. $\int_1^2 \frac{2x+1}{x+x^2} dx$<br>$= [\ln(x+x^2)]_1^2$<br>$= \ln(2+4) - \ln(1+1)$<br>$= \ln 6 - \ln 2$<br>$= \ln 3$                                           | A     |                   |

| MATHEMATICS: Question <i>m</i> choice                                                                                                                                                                                              |       |                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------|
| Suggested Solutions                                                                                                                                                                                                                | Marks | Marker's Comments |
| 4. $3x + 4y + 7 = 0$<br>$4y = -3x - 7$<br>$y = -\frac{3}{4}x - \frac{7}{4}$<br>$m_1 = -\frac{3}{4}$<br>$m_2 = \frac{4}{3}$<br>$y = mx + b$<br>$y = \frac{4}{3}x + b$<br>$3y = 4x + 3b$<br>$0 = 4x - 3y + 3b$<br>$0 = 8x - 6y + 6b$ | C     |                   |
| 5. $1 + 2m + 4m^2 + 8m^3 + \dots$<br>$r = 2m$<br>$-1 < 2m < 1$<br>$-\frac{1}{2} < m < \frac{1}{2}$                                                                                                                                 | C     |                   |

6  
MATHEMATICS: Question.....

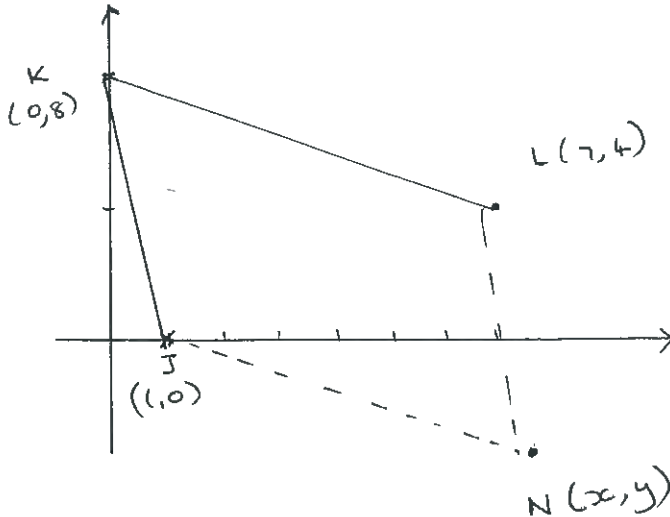
| Suggested Solutions                                                                                                                                                                                                                                                                                                                                         | Marks Awarded | Marker's Comments                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a) $48^\circ = \frac{48\pi}{180} = \frac{4\pi}{15}$ ✓                                                                                                                                                                                                                                                                                                       | 1 mark.       | Generally well done                                                                                                                                                        |
| b) (i) $m_{JL} = \frac{4-0}{7-1}$<br>$= \frac{4}{6}$<br>$= \frac{2}{3}$ ✓                                                                                                                                                                                                                                                                                   | 1 mark.       |                                                                                                                                                                            |
| (ii) $\tan \theta = \frac{2}{3}$<br>$\theta = 33^\circ 41' 24''$<br>$\theta = 34^\circ$ (nearest degree) ✓                                                                                                                                                                                                                                                  | 1 mark        | Question said to the nearest degree.                                                                                                                                       |
| (iii) Midpoint of JL<br>$\left( \frac{7+1}{2}, \frac{0+4}{2} \right) = (4, 2)$ ✓                                                                                                                                                                                                                                                                            | 1 mark        |                                                                                                                                                                            |
| (iv) $M_{KM} = \frac{8-2}{0-4} = \frac{+6}{-4} = -\frac{3}{2}$ ✓<br>$M_{JL} \times M_{KM} = \frac{2}{3} \times -\frac{3}{2}$<br>$= -1$<br>Products of gradients is -1 ✓<br>$\therefore KM \perp JL$                                                                                                                                                         | 2 marks       | 1 mark for getting correct gradient of KM<br>1 mark for proving products of gradients is -1.                                                                               |
| (v) Area $= \frac{1}{2} \times d_{JL} \times d_{KM}$<br>$= \frac{1}{2} \times \sqrt{(7-1)^2 + (4-0)^2} \times \sqrt{(0-4)^2 + (8-2)^2}$ ✓<br>$= \frac{1}{2} \sqrt{36+16} \times \sqrt{16+36}$<br>$= \frac{1}{2} \sqrt{52} \times \sqrt{52}$<br>$= \frac{1}{2} \times 2\sqrt{13} \times 2\sqrt{13}$<br>$= 26$ ✓<br>$\therefore$ Area is 26 unit <sup>2</sup> | 2 marks       | <br>1 mark - Distance JL and perpendicular height<br>1 mark - formula $\frac{1}{2}bh$ |

Suggested Solutions

Marks Awarded

Marker's Comments

b (vi)



For rhombus

Distance of JN = Distance of LN

Let N(x,y)

$$\therefore \sqrt{(x-1)^2 + (y-0)^2} = \sqrt{(x-7)^2 + (y-4)^2}$$

$$x^2 - 14x + 49 + y^2 - 8y + 16 = x^2 - 2x + 1 + y^2$$

$$0 = 12x + 8y - 64 \rightarrow \text{eq ①}$$

Now grad<sub>LN</sub> = grad<sub>KJ</sub>

$$\frac{y-4}{x-7} = \frac{8-0}{0-1}$$

$$\frac{y-4}{x-7} = -8$$

$$-8x + 56 = y - 4$$

$$y = -8x + 60 \quad \text{--- eq ②}$$

Sub eq ② in ①

$$12x + 8(-8x + 60) - 64 = 0$$

$$12x - 64x + 480 - 64 = 0$$

$$-52x + 416 = 0$$

$$52x = 416$$

$$x = 8$$

$$\text{when } x = 8 \Rightarrow y = 60 - 64 = -4$$

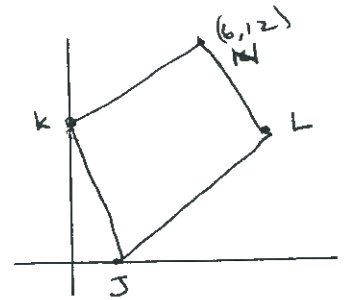
$$\therefore N \text{ is } (8, -4)$$

2 Marks

1 mark for working and getting 8

1 mark for substitution to get  $y = -4$ .

Few students got (6,12) as answer.



You did

JKNL and not JKLN

Maximum marks awarded was one.

Suggested Solutions

Marks Awarded

Marker's Comments

(c)

$$\frac{a}{2} = \frac{3}{32} \cdot 2^{n-2}$$

$$48 = 2^{n-2}$$

$$\ln 48 = n-2 \ln 2$$

$$n-2 = \ln 48 \div \ln 2$$

$$n = 5.585$$

$$\therefore n = 7.8587$$

Since  $n$  is not an integer  
 $\frac{a}{2}$  is not in the sequence ✓

2 marks

1 mark for  
substitution  
and  
manipulation

1 mark for  
explanation

OR

$$\frac{a}{2} = \frac{3}{2} (2^{n-2})$$

$$3 = \frac{1}{16} \times 2^{n-2}$$

$$3 = 2^{n-6}$$

but 2 to no  
power will give 3  
as an answer.

\* Few students  
used  $2^{n-2}$   
instead of  $2^{n-2}$ .

(d)(i)  $14 + 11 + 8 + 5 \dots$

$$a = 14 \text{ \& } d = -3 \quad \checkmark$$

$$\therefore T_n = a + (n-1)d$$

$$= 14 + (n-1)-3$$

$$= 14 - 3n + 3$$

$$= 17 - 3n \quad \checkmark$$

2 marks

1 for correct  
substitution  
of  $a$  and  
 $d$

1 mark for  
substitution  
and  
simplifying

(ii)  $14 + 11 + 8 + 5 \dots -34$

$$-34 = 17 - 3n$$

$$3n = 51$$

$$n = 17 \quad \checkmark$$

$$\therefore \sum_{n=1}^{17} (17 - 3n) \quad \checkmark$$

2 marks

1 mark for  
getting  
 $n = 17$

1 mark  
for  
summation

accepted

$$\sum_{n=1}^{17} (14 - 3n + 3)$$

as CFE

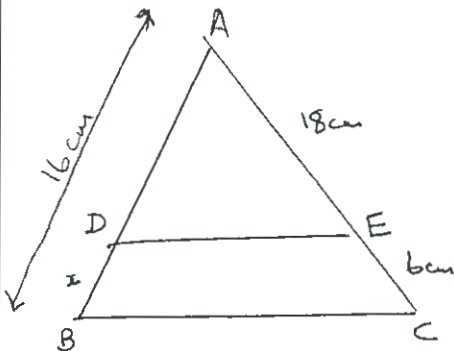
| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks Awarded  | Marker's Comments                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(e) <math>\log_b 5x^2</math>, <math>\log_b 5x</math>, <math>\log_b 5</math></p> <p><math>(\log_b 5 + 2\log_b x)</math>, <math>(\log_b 5 + \log_b x)</math>, <math>\log_b 5</math></p> <p>It is a AP with</p> $a = \log_b 5x^2$ $d = -\log_b x \quad \checkmark$ $T(n) = a + (n-1)d$ $= \log_b 5x^2 + (n-1)(-\log_b x) \quad \checkmark$ $= \log_b 5x^2 - n\log_b x + \log_b x$ $= \log_b 5 + 2\log_b x - n\log_b x + \log_b x$ $= \log_b 5 + 3\log_b x - n\log_b x$ $= \log_b 5 + \log_b x (3-n)$ $= (3-n)\log_b 5x \quad \checkmark$ <p>OR <math>\log_b 5x^{(3-n)}</math></p> | <p>3 marks</p> | <p>Poor attempt.</p> <p>1 mark for recognising <math>d = -\log_b x</math></p> <p>1 mark for substituting in <math>T_n = a + (n-1)d</math></p> <p>1 mark for final answer.</p> |
| <p>(f) <math>S_n = n^2 - 4n</math></p> $S_{n-1} = (n-1)^2 - 4(n-1)$ $= n^2 - 2n + 1 - 4n + 4$ $= n^2 - 6n + 5 \quad \checkmark$ $T_n = S_n - S_{n-1} \quad \checkmark$ $n^2 - 4n - (n^2 - 6n + 5)$ $n^2 - 4n - n^2 + 6n - 5$ $2n - 5$                                                                                                                                                                                                                                                                                                                                             | <p>2 marks</p> | <p>Mostly well done</p> <p>1 mark for getting <math>S_{n-1}</math></p> <p>1 mark for manipulating <math>T_n</math>.</p>                                                       |

Suggested Solutions

Marks Awarded

Marker's Comments

g)



(i) Prove  $\triangle ADE \parallel \triangle ABC$

In  $\triangle ADE, \triangle ABC$

$\hat{A}$  is common

$\hat{AED} = \hat{ACB}$  (Corresponding angles are equal as  $DE \parallel BC$ )

$\therefore \triangle ADE \parallel \triangle ABC$  (equiangular)

(ii) length of DB

$\frac{AD}{AB} = \frac{AE}{AC}$  (Corresponding sides in similar triangles are in the same ratio)

$$\frac{16-x}{16} = \frac{18}{24}$$

$$16-x = \frac{288}{24}$$

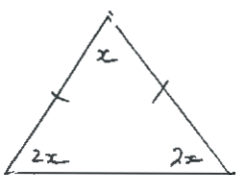
$$x = 4$$

$$\therefore DB = 4 \text{ cm.}$$

4 marks

Question generally very well done

h)



let vertical angle =  $x$

$$2x + 2x + x = 180^\circ \text{ (angle sum of triangle)}$$

$$5x = 180$$

$$x = 36$$

$\therefore$  Angles are

$$36^\circ, 72^\circ \text{ and } 72^\circ$$

2 marks

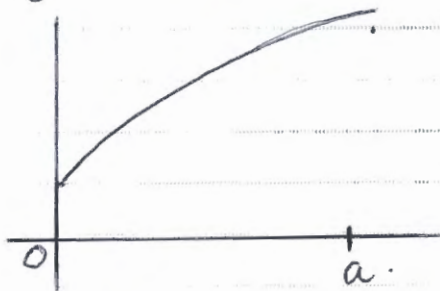
One had to write all three angles to get full marks.

Suggested Solutions

Marks

Marker's Comments

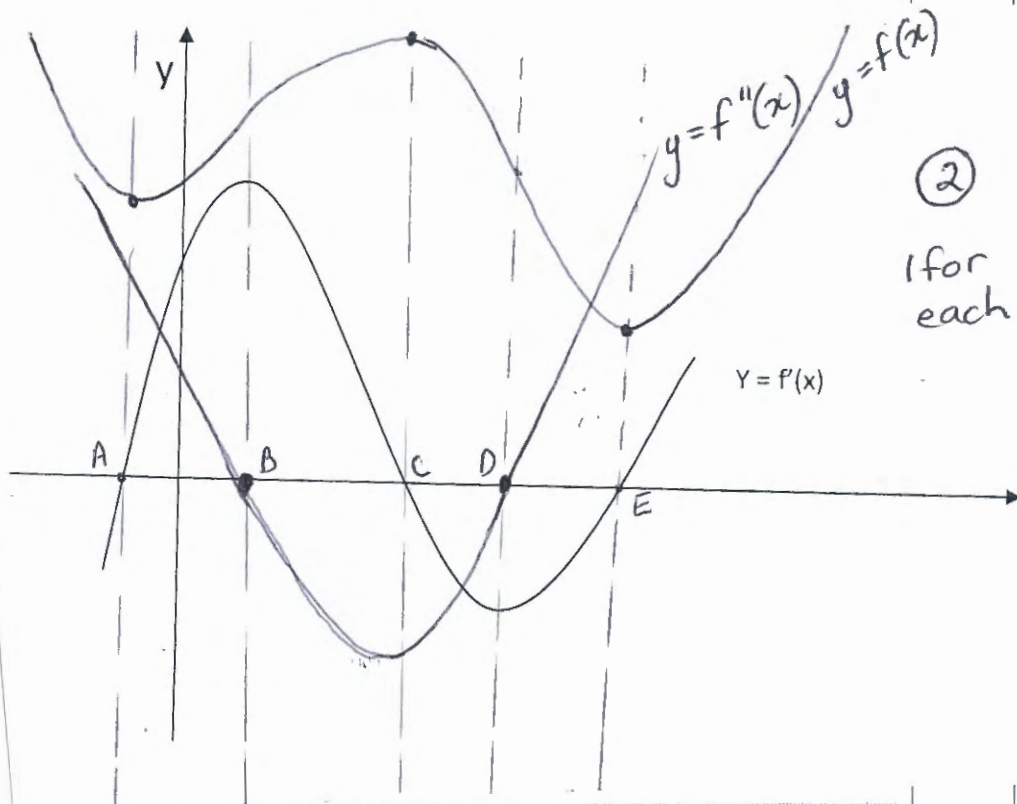
- (a)  $y'' < 0$  concave down  
 $y' > 0$  increasing  
 $y > 0$  above  $x$  axis



①

No half marks.  
 Needed to be  
 concave down  
 + increasing.  
 + above axis.

(b)



②

1 for each.

Look at important features to find  $f(x)$ .

At A:  $f'(x) = 0$  &  $f''(x) > 0$   
 $\therefore$  min TP

At B:  $f''(x) = 0$  &  $f'(x) > 0$   
 changes sign  $\therefore$  POI

At C:  $f'(x) = 0$  &  $f''(x) < 0$   
 $\therefore$  max TP

At D:  $f''(x) = 0$  &  $f'(x) < 0$   
 changes sign  $\therefore$  POI

At E:  $f'(x) = 0$  &  $f''(x) > 0$   
 $\therefore$  min TP.

(c)  $y = x^3 - 12x + 5$   
 $y' = 3x^2 - 12$   
 $y'' = 6x$

Possible stationary point when  $y' = 0$   
 ie  $3x^2 - 12 = 0$   
 $x^2 - 4 = 0$   
 $x = \pm 2$

①

Suggested Solutions

Marks

Marker's Comments

When  $x=2$   $y=8-24+5$   
 $=-11$

$x=-2$   $y=-8+24+5$   
 $=21$

∴ Stationary points at  $(2, -11)$  and  $(-2, 21)$  ①

To test the nature of the stationary point look at  $y''$

When  $x=2$   $y''=12 > 0$  ∴ concave up  
 ∴ (relative) min TP at  $(2, -11)$  ①

When  $x=-2$   $y''=-12 < 0$  ∴ concave down  
 ∴ (relative) max TP at  $(-2, 21)$  ①

(ii) Point of inflexion when  $y''=0$

AND concavity changes.

ie  $6x=0$   
 $x=0$

|       |       |   |       |
|-------|-------|---|-------|
| $x$   | -1    | 0 | 1     |
| $y''$ | -6    | 0 | 6     |
|       | $< 0$ |   | $> 0$ |

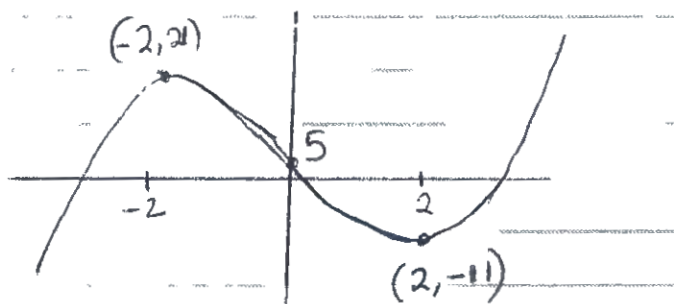
∴ Change in concavity

When  $x=0$   $y=5$

∴ Point of Inflexion at  $(0, 5)$

① for  $y''=0$  ∴  $x=0$   
 $(0, 5)$

① for checking



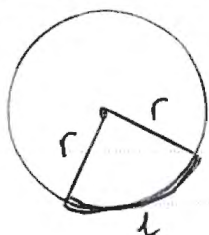
①

Suggested Solutions

Marks

Marker's Comments

(d)



$$\text{Area sector} = 96 \text{ cm}^2$$

$$A = \frac{1}{2} r^2 \theta = 96 \text{ cm}^2$$

$$\frac{1}{2} r^2 \theta = 96$$

$$r^2 \theta = 192 \quad \text{--- ①}$$

$$\theta = \frac{192}{r^2}$$

$$\text{Perimeter} = 56 \text{ cm}$$

$$\text{ie } l + 2r = 56 \quad \text{where } l = r\theta$$

$$r\theta + 2r = 56$$

$$r(\theta + 2) = 56 \quad \text{--- ②}$$

$$\text{Subst } \theta = \frac{192}{r^2} \quad \text{--- from ①}$$

$$r\left(\frac{192}{r^2} + 2\right) = 56$$

$$\frac{192}{r} + 2r = 56$$

$$192 + 2r^2 = 56r$$

$$r^2 - 28r + 96 = 0$$

$$(r - 24)(r - 4) = 0$$

$$r = 24 \quad \text{OR} \quad r = 4$$

$$\text{When } r = 4 \quad \theta = \frac{192}{r^2} = 12^\circ$$

$$\text{When } r = 24 \quad \theta = \frac{192}{24^2} = \frac{1}{3}$$

However,  $0 \leq \theta \leq 2\pi$  because it is a sector of circle.  $\therefore \theta \neq \frac{1}{3} > 2\pi$   
 $\therefore \theta = \frac{1}{3}^\circ$  only.

$$\theta = \frac{180}{\pi} \times \frac{1}{3} = 19.0986$$

$$\therefore \theta = 19^\circ \quad \text{and} \quad r = 24$$

①

①

①

①

①

Read all the question. You were asked for  $\theta$  to the nearest degree.

Suggested Solutions

Marks

Marker's Comments

(e) Let the smaller part be  $x$ , then the larger part is  $36-x$ .

$$P = x(36-x)^2$$

$$P = x(1296 - 72x + x^2)$$

$$P = 1296x - 72x^2 + x^3$$

$$\frac{dP}{dx} = 1296 - 144x + 3x^2$$

(ii) Possible maximum product when

$$\frac{dP}{dx} = 0$$

$$\text{ie } 1296 - 144x + 3x^2 = 0$$

$$x^2 - 48x + 432 = 0$$

$$(x-12)(x-36) = 0$$

$$x = 12, x = 36 \quad x < 36$$

$$\therefore x = 12$$

$$\frac{d^2P}{dx^2} = -144 + 6x$$

$$\text{When } x = 12 \quad \frac{d^2P}{dx^2} = -144 + 6 \times 12 = -72$$

$\therefore$  concave down

$\therefore$  Max product occurs when

$$x = 12 \quad \text{ie } P = 12(36-12)^2 = 6912$$

$\therefore$  maximum product possible is 6912

Method important  
- you knew the answer.

Note! If you could not do (i) then use the given result to do (ii). Don't leave it out!!

(f)  $\frac{dy}{dx} = \sec^2 2x$

$$y = \frac{1}{2} \int 2 \sec^2 2x \, dx$$

$$= \frac{1}{2} (\tan 2x) + c$$

①

When  $y=0$   $x=\frac{\pi}{6}$

$$0 = \frac{1}{2} \tan \frac{2\pi}{6} + c$$

$$0 = \frac{1}{2} \sqrt{3} + c$$

$$c = -\frac{\sqrt{3}}{2}$$

i.  $y = \frac{1}{2} \tan 2x - \frac{\sqrt{3}}{2}$

①

(g)  $y = 4 \cos 2x$

$$A = 2 \int_0^{\frac{\pi}{6}} 4 \cos 2x \, dx$$

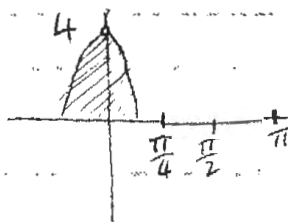
$$= \frac{8}{2} \int_0^{\frac{\pi}{6}} 2 \cos 2x \, dx$$

$$= 4 \left[ \sin 2x \right]_0^{\frac{\pi}{6}}$$

$$= 4 \sin \frac{2\pi}{6} - 4 \sin 0$$

$$= 4 \times \frac{\sqrt{3}}{2} - 4 \times 0$$

$$= 2\sqrt{3} u^2$$



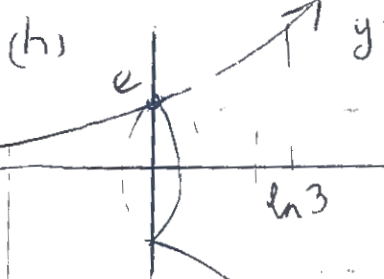
①

①

Suggested Solutions

Marks

Marker's Comments



$$V = \pi \int_0^{\ln 3} (e^{x+1})^2 dx$$

$$= \pi \int_0^{\ln 3} e^{2x+2} dx$$

$$= \pi \left[ \frac{e^{2x+2}}{2} \right]_0^{\ln 3}$$

$$= \pi \left[ \frac{e^{2\ln 3+2}}{2} - \frac{e^2}{2} \right]$$

$$= \frac{\pi}{2} (e^{2\ln 3} \cdot e^2 - e^2)$$

$$= \frac{\pi}{2} (e^{\ln 9} e^2 - e^2)$$

$$= \frac{\pi}{2} (9e^2 - e^2)$$

$$= \frac{8e^2\pi}{2}$$

$$= 4e^2\pi u^3$$

$$= 92.85361743$$

$$= 92.85 u^3 \text{ (2dp)}$$

Volume  
Formula wrong  
0 marks!

Note:

①  $(e^{x+1})^2 \neq (e^x+1)^2$

①

①

Note if you had  
CFE, the mark  
for 2dp was not  
given unless you  
had the full calculator  
answer first.

SUGGESTED SOLUTIONS.

MARKS

AWARDED

MARKERS

COMMENT

QUESTION 8)

ai)

$$y = \ln \sqrt{\frac{1+x}{1-x}}$$

$$= \frac{1}{2} \ln(1+x) - \frac{1}{2} \ln(1-x) \quad \text{--- (1)}$$

$$\frac{dy}{dx} = \frac{1}{2} \cdot \frac{1}{1+x} - \frac{1}{2} \cdot \frac{1}{1-x} \quad \cdot -1$$

$$= \frac{1}{2(1+x)} + \frac{1}{2(1-x)}$$

$$= \frac{1(1-x) + (1+x)}{2(1-x)^2} \quad \text{--- (1)}$$

$$= \frac{1-x+1+x}{2(1-x)^2} = \frac{2}{2(1-x)^2}$$

$$= \frac{1}{(1-x)^2} \quad \text{(shown)}$$

common  
mistake  
not multiplying  
(-1)

ii)

$$\int_0^{1/3} \frac{4dx}{(1-x)^2}$$

$$= 4 \int_0^{1/3} \frac{1}{(1-x)^2} dx$$

$$= 4 \left[ \ln \sqrt{\frac{1+x}{1-x}} \right]_0^{1/3} \quad \text{--- (1)}$$

$$= 4 \ln \sqrt{\frac{2 \frac{4}{3} \times \frac{3}{2}}{\frac{1}{3}}} - \ln \sqrt{\frac{1}{1}}$$

$$= 4(\ln \sqrt{2} - \ln \sqrt{1})$$

$$= 4 \ln \sqrt{2}$$

$$\text{or } 4 \ln 2^{\frac{1}{2}}$$

$$= \ln 2^2 = 2 \ln 2 \quad \text{--- (1)}$$

SUGGESTED SOLUTIONS.

MARKS

AWARDED

MARKERS

COMMENT

8b)

$$\int_0^{\pi/8} (1 - \cot 4x) dx$$

$$\int_0^{\pi/8} 1 - \frac{\cos 4x}{\sin 4x} dx$$

$$\left[ x - \frac{1}{4} \ln(\sin 4x) \right]_0^{\pi/8}$$

$$= \left[ \frac{\pi}{8} - \frac{1}{4} \ln(\sin \frac{\pi}{2}) \right] - \left[ \left( -\frac{1}{4} \ln \sin 4(0) \right) \right]$$

$$= \frac{\pi}{8} - \frac{1}{4} (0) + \text{undefined.}$$

$$= \frac{\pi}{8}$$

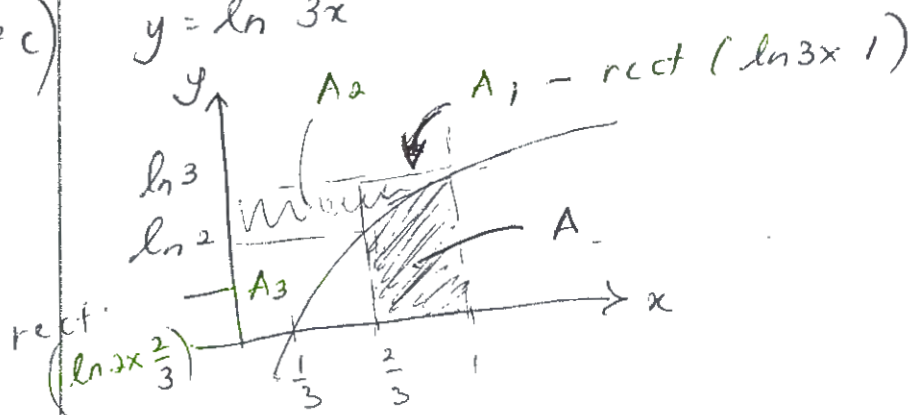
(1)

many students didn't write undefined but instead wrote 0 which is incorrect.

(1)

8c)

$$y = \ln 3x$$



$$\begin{aligned} A_2 &= \int_{\ln 2}^{\ln 3} \frac{1}{3} e^y \\ &= \frac{1}{3} \left[ e^y \right]_{\ln 2}^{\ln 3} \\ &= \frac{1}{3} (e^{\ln 3} - e^{\ln 2}) \\ &= \frac{1}{3} (1) = \frac{1}{3} \end{aligned}$$

(1)

SUGGESTED SOLUTIONS.

MARKS  
AWARDED

MARKERS  
COMMENT

$$\begin{aligned} \therefore A &= A_1 - A_3 - A_2 \\ &= \ln 3 - \frac{2}{3} \ln 2 - \frac{1}{3} \\ &= \frac{1}{3} (3 \ln 3 - 2 \ln 2 - 1) u^2 \end{aligned}$$

①

①

to find  
 $A_1$  &  $A_3$

8d)  $y = x \sin x$  .  $y^2 = x^2 \sin^2 x$ .  
for Simpson's rule from  $0 - \frac{\pi}{2}$

| x     | 0 | $\frac{\pi}{8}$                        | $\frac{\pi}{4}$                        | $\frac{3\pi}{8}$                         | $\frac{\pi}{2}$                        |
|-------|---|----------------------------------------|----------------------------------------|------------------------------------------|----------------------------------------|
| $y^2$ | 0 | $\frac{\pi^2}{8} \sin^2 \frac{\pi}{8}$ | $\frac{\pi^2}{4} \sin^2 \frac{\pi}{4}$ | $\frac{3\pi^2}{8} \sin^2 \frac{3\pi}{8}$ | $\frac{\pi^2}{2} \sin^2 \frac{\pi}{2}$ |
| w     | 1 | 4                                      | 2                                      | 4                                        | 1                                      |
|       | 0 | 0.02258                                | 0.30842                                | 1.18466                                  | 2.4674                                 |

①

able to  
write weighting  
& intervals  
correctly.

$$\therefore V = \pi \int_0^{\pi/2} x^2 \sin^2 x \, dx$$

using Simpson's Rule

$$= \pi \left[ \frac{\frac{\pi}{8} - 0}{3} (0 + 0.02258 \times 4 \right.$$

$$\left. + 0.30842 \times 2 + 1.18466 \times 4 + 2.4674) \right]$$

①

h correct.

$$= \pi \left( \frac{\pi}{24} \right) (7.91321 \dots)$$

$$= 3.25 u^3 \text{ (2 dp)}$$

①

to write  
V with  $\pi$   
& y is squared

SUGGESTED SOLUTIONS.

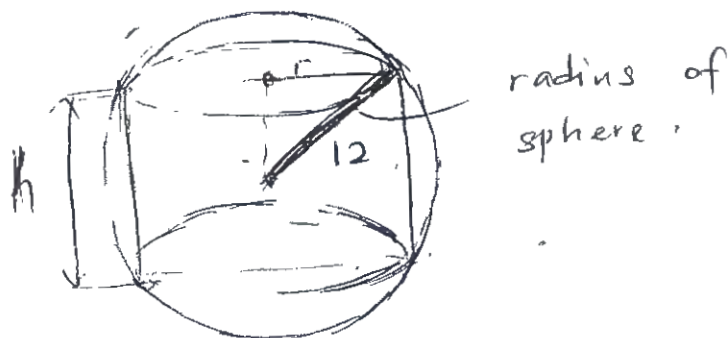
MARKS

AWARDED

MARKERS

COMMENT

8e) i)



using pythagoras

$$h^2 + 2r^2 = 24^2 \quad \text{OR} \quad r^2 + \left(\frac{h}{2}\right)^2 = 12^2$$

$$4r^2 = 576 - h^2$$

$$r = \sqrt{\frac{576 - h^2}{4}}$$

$$= \frac{\sqrt{576 - h^2}}{2} \quad r > 0$$

①

$$V = \pi h r^2$$

$$= \pi h \left( \frac{576 - h^2}{4} \right)$$

} ①

$$= \frac{\pi h}{4} (576 - h^2) \text{ shown}$$

ii)

To find max

do  $\frac{dV}{dh}$  and expand  $V$  from above

$$\therefore V = 144\pi h - \frac{\pi h^3}{4}$$

SUGGESTED SOLUTIONS.

MARKS

MARKERS

AWARDED

COMMENT

$$\frac{dv}{dh} = 144\pi - \frac{3\pi h^2}{4}$$

$$\frac{dv}{dh} = 0$$

$$\frac{3\pi h^2}{4} = 144\pi$$

$$h^2 = 192$$

$$h = 8\sqrt{3}$$

$$h > 0$$

(1)

$$\frac{d^2v}{dh^2} = -\frac{6\pi h}{4}$$

$$= -\frac{3\pi h}{2}$$

$$\text{when } h = 8\sqrt{3}$$

$$\frac{d^2v}{dh^2} < 0$$

(1)

$$\therefore \text{rel max at } h = 8\sqrt{3}$$

OR.

|                 |    |             |    |
|-----------------|----|-------------|----|
| $h$             | 12 | $8\sqrt{3}$ | 14 |
| $\frac{dv}{dh}$ | +  | 0           | -  |

$$\therefore V_{\max} = 144\pi(8\sqrt{3}) - \frac{\pi(8\sqrt{3})^3}{4}$$

$$= 1152\sqrt{3}\pi - 384\sqrt{3}\pi$$

$$= 768\sqrt{3}\pi \text{ u}^3$$

(1)

some students assumed it will be max without doing  $\frac{d^2v}{dh^2}$  nor the table no. marks awarded.

SUGGESTED SOLUTIONS.

MARKS  
AWARDED

MARKERS  
COMMENT

8e) iii) Ratio  $V_{\text{max cylinder}} : V_{\text{sphere}}$

$$V_{\text{sphere}} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \pi (12)^3$$

$$\therefore 768\sqrt{3}\pi : \frac{4}{3}\pi(1728)$$

$$\sqrt{3} : 3$$

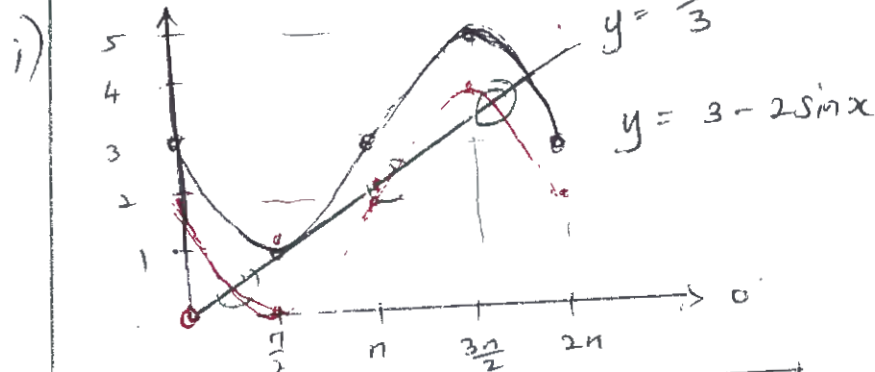
$$\text{or } 1 : \sqrt{3}$$

common  
error - students  
used  $r = 8\sqrt{3}$

any of this form  
is (1) mark.

no marks awarded  
if written in decimal

f) Graph  $y = 3 - 2\sin x$   $0 \leq x \leq 2\pi$



1 mark for shape  
1 mark for end points.

| $x$               | 0 | $\frac{\pi}{2}$ | $\pi$ | $\frac{3\pi}{2}$ | $2\pi$ |
|-------------------|---|-----------------|-------|------------------|--------|
| $2\sin x$         | 0 | 2               | 0     | -2               | 0      |
| $y = 3 - 2\sin x$ | 3 | 1               | 3     | 5                | 3      |

ii.

$$1 - \sin x - \frac{x}{3} = 0$$

$$(1 - \sin x = \frac{x}{3}) \times 2$$

$$2 - 2\sin x = \frac{2x}{3}$$

(1) mark  
shift the graph down  
from 3  $\rightarrow$  2 then draw  
 $y = \frac{2x}{3}$

SUGGESTED SOLUTIONS.

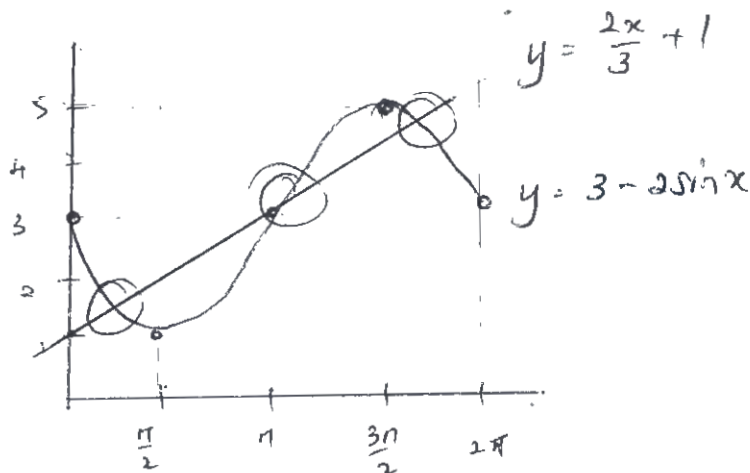
MARKS  
AWARDED

MARKERS  
COMMENT

OR.

$$2 - 2\sin x = \frac{2x}{3}$$

$$3 - 2\sin x = \frac{2x}{3} + 1$$



$\therefore$  3 solutions

— (1) mark

89)

$$A_1 = 10000 \times \frac{241}{240}$$

$$A_2 = (A_1 + m) \times \frac{241}{240}$$

$$= 10000 \left( \frac{241}{240} \right)^2 + m \left( \frac{241}{240} \right)$$

but it's  
compounded  
 $\therefore r = \frac{241}{240}$

$$A_3 = 10000 \left( \frac{241}{240} \right)^3 + m \left( \frac{241}{240} \right)^2 + m \left( \frac{241}{240} \right)$$

$$A_{60} = 10000 \left( \frac{241}{240} \right)^{60} + m \left( \frac{241}{240} \right)^{59} + \dots + m \left( \frac{241}{240} \right)$$

$$= 10000 \left( \frac{241}{240} \right)^{60} + m \left[ \left( \frac{241}{240} \right) + \left( \frac{241}{240} \right)^2 + \dots + \left( \frac{241}{240} \right)^{59} \right]$$

$$160000 = 10000 \left( \frac{241}{240} \right)^{60} + m \left[ \frac{\frac{241}{240} \left( \frac{241}{240}^{59} - 1 \right)}{\frac{1}{240}} \right]$$

$$m = \$21\,96.31 \quad (\text{nearest cent})$$

(1) mark  
for writing  
 $A_1, A_2 \rightarrow A_{60}$

(1) mark to  
write in GP  
form

— (1)

| SUGGESTED SOLUTIONS.                                                                                                                                                                                                                                                                                                                                                           | MARKS<br>AWARDED                                             | MARKERS<br>COMMENT |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------|
| <p>OR.</p> $A_1 = 10000 R.$ $A_2 = 10000 R^2 + MR$ $A_3 = 10000 R^3 + MR^2 + MR.$ $A_n = 10000 R^n + M(R + R^2 + \dots + R^{n-1})$ $= 10000 R^n + M \frac{R(R^{n-1} - 1)}{R - 1}$ $A_{60} = 10000 \left(\frac{241}{240}\right)^{60} + M \left( \frac{\frac{241}{240} \left( \frac{241}{240}^{59} - 1 \right)}{\frac{241}{240} - 1} \right)$ $= \$2196.31 \quad (2 \text{ dp})$ | $a = R = \frac{241}{240}$ $R = R = \frac{241}{240}$ $n = 60$ |                    |